Proprietary to General Electric Company
Direction 5438996-8EN, **Rev#** 9
Brivo CT385 Series Service Methods (Adv)
Module Version 1.0, **Version Date** 1/14/13

ISO Alignment Process

(Click Here for Summary Version)

1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	Not Applicable	Not Applicable

2 Overview

Cold ISO Alignment is done before the tube is heated by Generator calibrations. Hot ISO is done later.

Illustration 1: ISO Flowchart

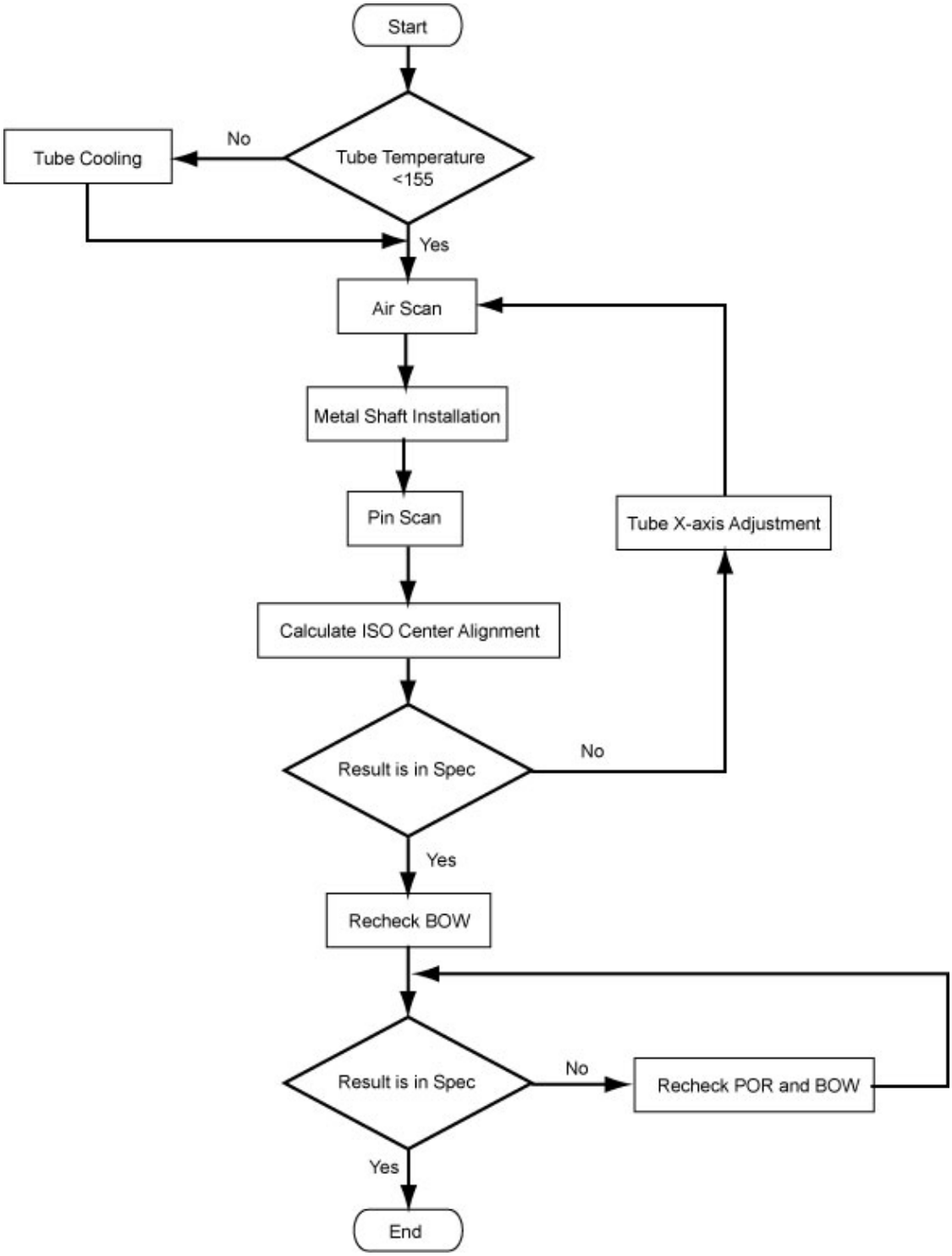
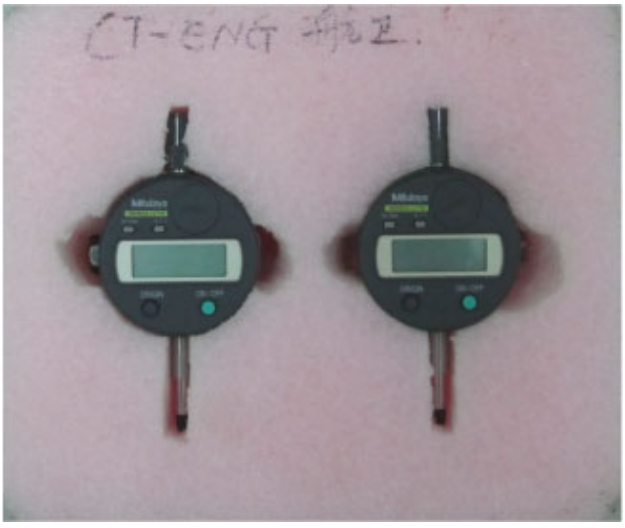


Illustration 2: Dial Gauge Indicator



3 Preliminary Requirements

3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
Dial Gauge Indicator (millimeters or inches) - see Illustration 2	2	-	-	-

3.2 Safety



⚠ CAUTION

No personnel or patients should be in the area of the gantry while scanning without adequate radiation safety precautions being utilized.

3.3 Required Conditions

Condition	Reference	Effectivity
Recommend procedure completion by GE Service trained personnel.	-	-
The positioning lights must be OK before ISO alignment.	-	-

4 Procedure

4.1 For Tube Change Only

Wait 90 minutes if the new tube had more than 25 Kilo Joules of energy input, [KV x mA x Sec ÷ 1000] within the last 30 minutes prior to the start of system alignments. If a tube heat soak has been performed you must wait a minimum of 6 hours before system alignments can be performed.

4.2 Confirm Tube Cooling

When performing alignments, wait at least 15-30 minutes between scans to prevent unnecessary adjustments.

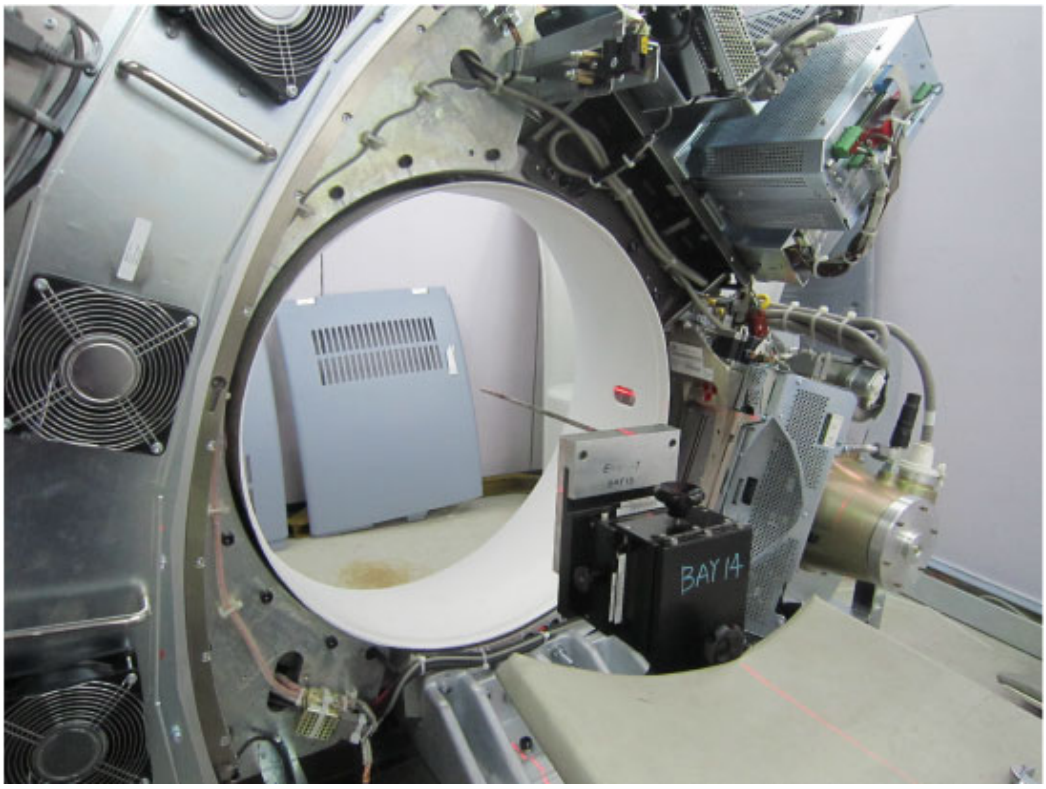
4.3 Preparation

1. Move the cradle OUT to its fully retracted position (farthest from gantry) and position the table to its lowest height.
2. Remove the gantry left side cover.
3. Turn OFF “Axial Enable”, “HVDC” and “120VAC” switches on the service switch panel.
4. Remove the table foot switch cover.
5. Remove the gantry covers as required.
6. Turn ON “HVDC”, “Axial Enable” and “120VAC” switches.

4.4 Adjustment Procedure

1. You should be familiar with *Plane of Rotation (POR) Alignment Process* before proceeding with ISO alignment.
2. Select [Service] ->[Calibration] -> [ISO Alignment] to enter ‘ISO feature’ UI.
3. Click [Scan] to start **Execute air scans**.
4. Hint information ‘ISO air scans were successful.’ displays on the screen.
5. Elevate the table to 135mm height.
6. Insert the phantom holder into the cradle and attach a round, 0.125 inch diameter metal shaft to it.
7. Align the localization light on the metal shaft.

Illustration 3: Position Metal Shaft



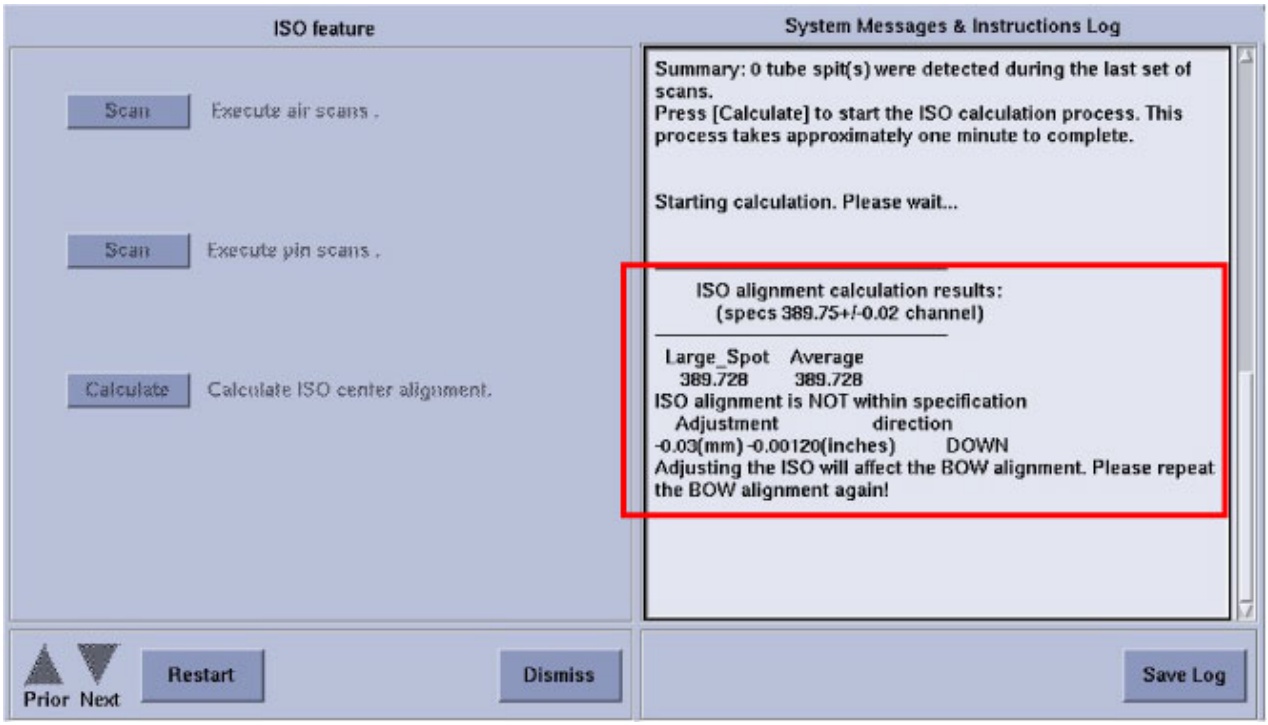
8. Press [External Landmark] button to position the positioning light to the **190**.

Illustration 4: External Landmark



9. Press [Cradle In] button to move the cradle until **0** displays on the display/control panel.
10. Click [Scan] to start **Execute pin scans**.
11. Click [Calculate] to start **Calculate ISO center alignment**, and the account result displays on the monitor.

Illustration 5: Account Result



12. Adjust the Tube X-axis position.

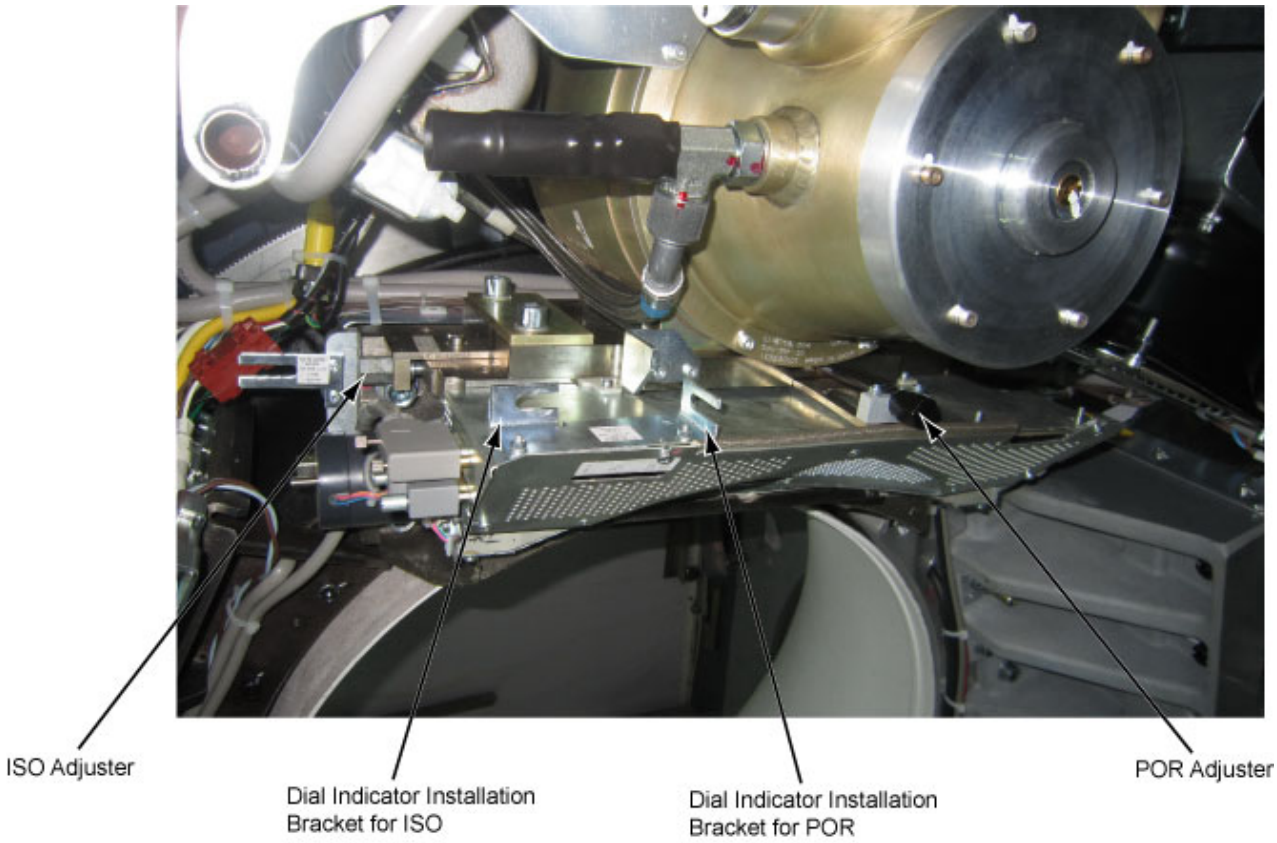
    a. Rotate the gantry until the tube reaches about 3 o'clock position.

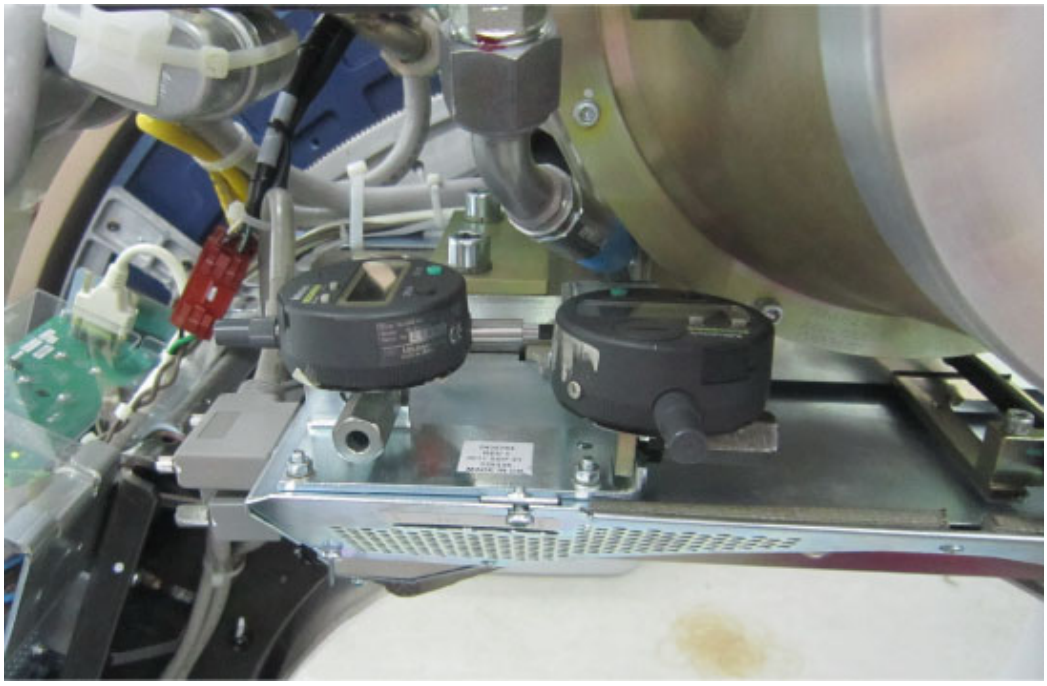
    b. Turn OFF "HVDC", "Axial Enable" and "120VAC" switches on the service switch panel.

    c. Mount the two dial indicators onto the collimator / tube assembly - one for ISO alignment and one for POR. Make sure that you zero both of them (POR and ISO).

    NOTE: The dial indicators must be vertical with the gauge adjustment surface.

Illustration 6: Dial Indicators Position





d. Loosen the four M12 mounting bolts that secure the tube. (1/2 turn out is all that's necessary, and not any more.)

Illustration 7: Tube Mounting Bolts



e. Adjust the tube position in X-axis by turning the ISO adjuster according to the calculative result displayed on the monitor screen.

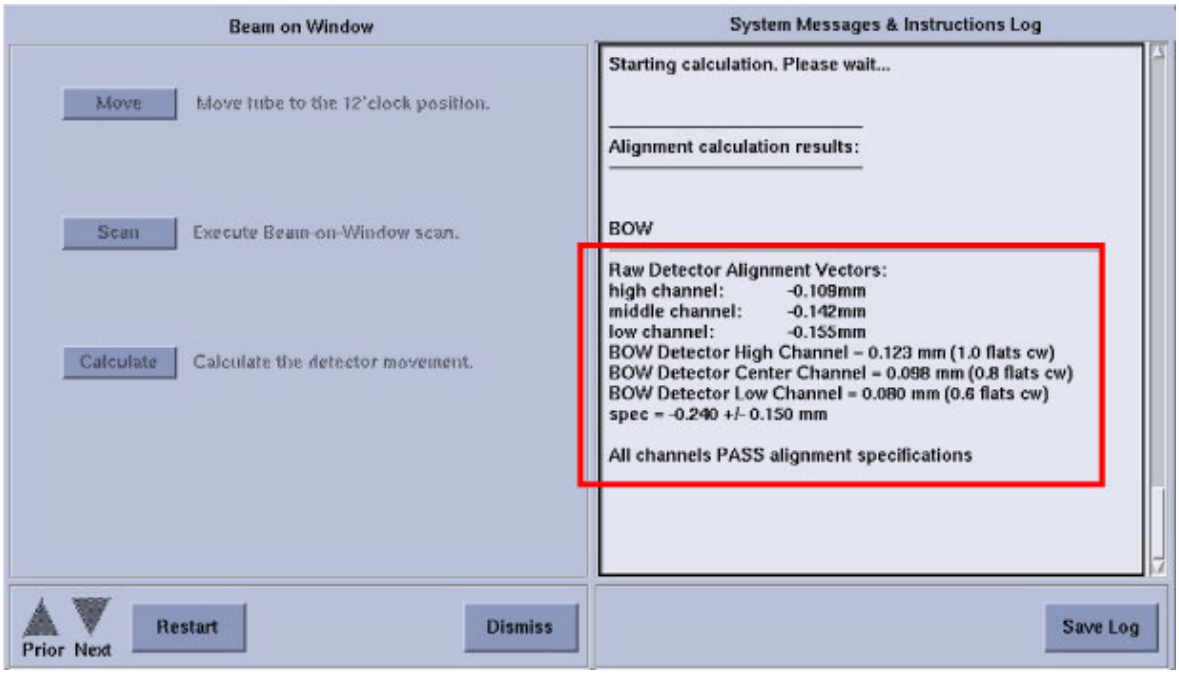
   While moving the tube in the ISO direction the tube will likely move POR out of alignment. By watching both dial indicators you can keep POR in alignment while moving ISO. Go slow - being accurate while moving the tube will reduce the number of iterations, because no unnecessary heat will be put into the tube.

13. Secure M12 mounting bolts to pre-load torque specification as shown below.

Table 1: Pre-load Torque Specification

25 ft.-lbs	33 N-m	295 in-lbs	340 kg-cm

14. Turn ON “Axial Enable”, “HVDC” and “120VAC” switches.
15. Click [Restart] to repeat ISO procedure from Step 3 to Step 11.
16. Click [Save Log], then click [Dismiss] to quit ISO.
17. Wait 15 minutes, then recheck BOW to verify values are within specification -0.240mm ± 0.150mm.
    - If BOW is within spec, DO NOT adjust anything



    - If BOW is out of spec, alignments must be done, starting with POR and ending with BOW recheck.
18. Apply final torque on all four M12 mounting bolts according to the values specified in Table 2.

Table 2: Final Torque

49 ft.-lbs	66 N-m	590 in-lbs	680 kg-cm

19. Restore the gantry to original configuration and reinstall the table foot switch cover.

5 Finalization

No finalization steps.

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